

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (3 Questions)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	30 (3 Questions × 10 Marks)
CO5 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools (BL6)</i>		
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Section / Batch:	BTECH AI & DS	Date:	18-08-2026

Code of Conduct

I Renee Vincent ASR-192424161 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 3 questions, each carrying 10 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Visualizing Uncertainty	Probability visualization, confidence intervals, uncertainty interpretation	1	1
Handling Overplotting	Overplotting, transparency, jittering, alternative plots	2	1
Principles of Effective Visualization & Accessibility	Proportional ink principle, misleading axes, accessible color design	3	1
GRAND TOTAL:			30 Marks

Annexure A – Scenario Based Assessment

1. A weather forecasting agency predicts rainfall probability (%) for a week along with uncertainty:

Day	Rainfall Probability	Confidence Interval
Mon	60	10
Tue	65	20
Wed	70	5
Thu	55	15
Fri	80	25

As a data analyst, design a visualization that clearly communicates both probability and uncertainty. Which day's prediction is most reliable? Justify your answer using visualization principles.

Aim:

To visualize the predicted rainfall probability along with its uncertainty for different days of the week and identify the day with the most reliable prediction using an **error-bar plot**.

Suitable Graph:

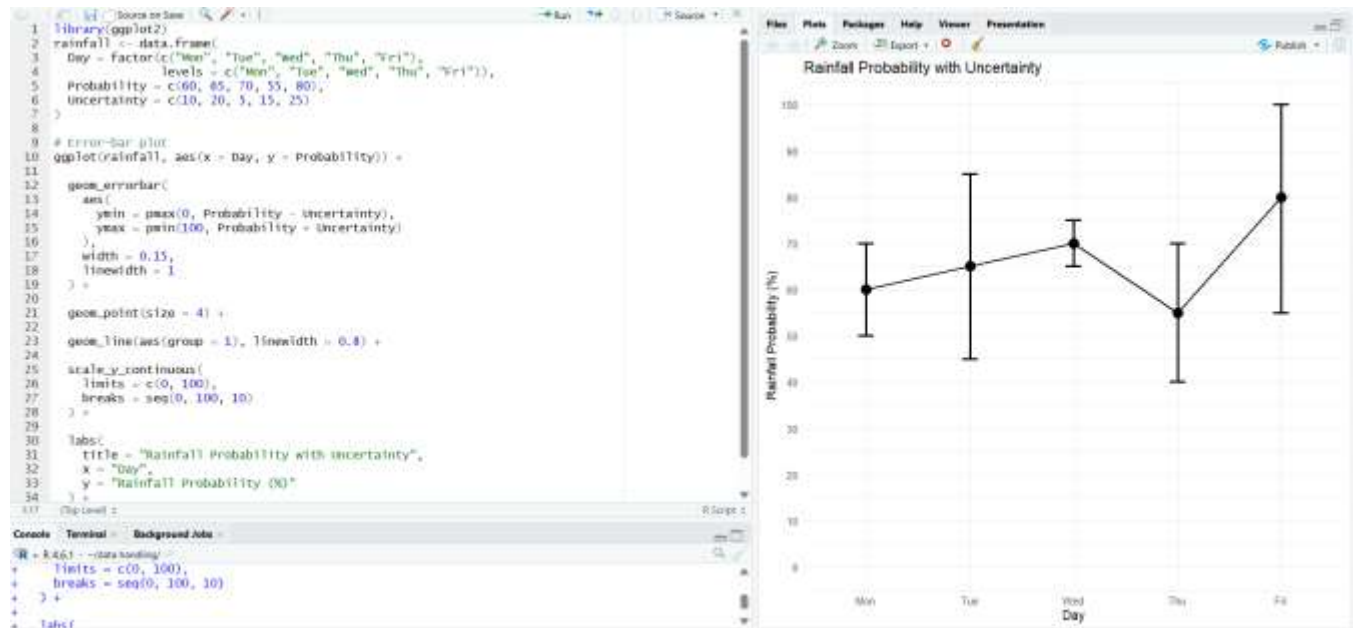
Error-Bar Plot with Points and Line

An error bar plot is suitable because it shows the rainfall probability and its uncertainty together. The error-bar length makes it easy to compare the reliability of each day's prediction.

Procedure:

- Open **RStudio** and create a new R script.
- Load the **ggplot2** package.
- Create a data frame containing day, rainfall probability, and uncertainty.
- Use `geom_point()` for probability and `geom_errorbar()` for uncertainty.
- Run the program and compare the lengths of the error bars.

Output:



Justification:

Wednesday is the most reliable prediction because its uncertainty is only $\pm 5\%$, which is the smallest among all days. Its narrow error bar indicates greater precision. Error bars communicate both the predicted value and the associated uncertainty without hiding either information.

2. A mobile app company collects user ratings (1–5) from 500 users. Many users give the same ratings:

Rating	Frequency
1	50
2	80
3	150
4	140
5	80

If plotted directly, the visualization suffers from overplotting. Propose a suitable visualization technique and explain how transparency or jittering can improve clarity. Suggest an alternative plot if needed.

Aim:

To visualize the distribution of user ratings while reducing the problem of overplotting caused by repeated ratings.

Suitable Graph:

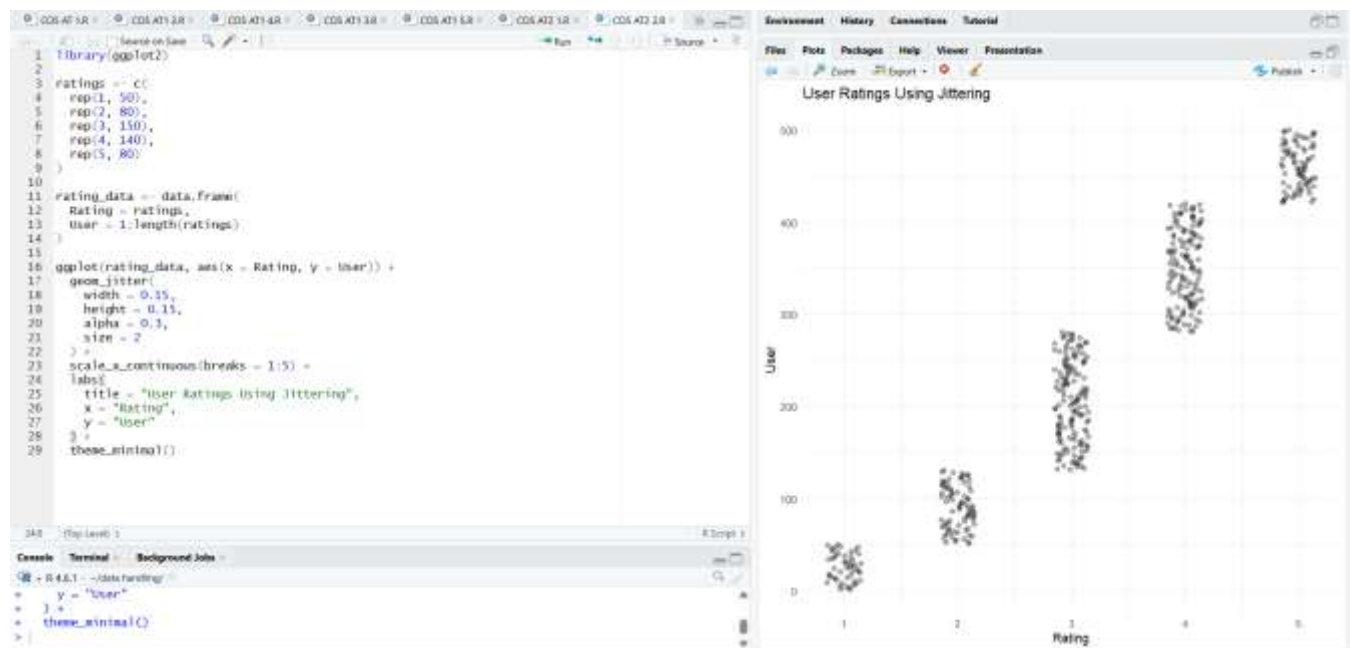
Jitter Plot with Transparency

An alternative is a **Bar Chart**, which is particularly suitable because the data is already given as frequencies.

Procedure:

- Open RStudio and load the ggplot2 package.
- Create individual observations using rep() according to the given frequencies.
- Use geom_jitter() to slightly separate overlapping observations.
- Set alpha = 0.3 to make overlapping points transparent.
- Run the program and observe the concentration of ratings.

Output:



Justification:

Jittering separates points that would otherwise overlap at the same rating. Transparency makes dense regions more visible because overlapping points appear darker. Rating 3 has the highest frequency (150 users). If only frequencies need to be compared, a bar chart is an even simpler alternative.

3. A company presents employee satisfaction scores:

Department	Score (%)
HR	70
Sales	75
IT	78
Finance	80

The chart uses a truncated y-axis (starting at 65) and red-green color coding. Identify the issues in this visualization. Redesign it using proper axis design, proportional ink principle, and accessible color choices. Explain how your design avoids misleading interpretation.

Aim:

To redesign the employee satisfaction visualization so that the differences between departments are represented accurately and accessibly.

Suitable Graph:

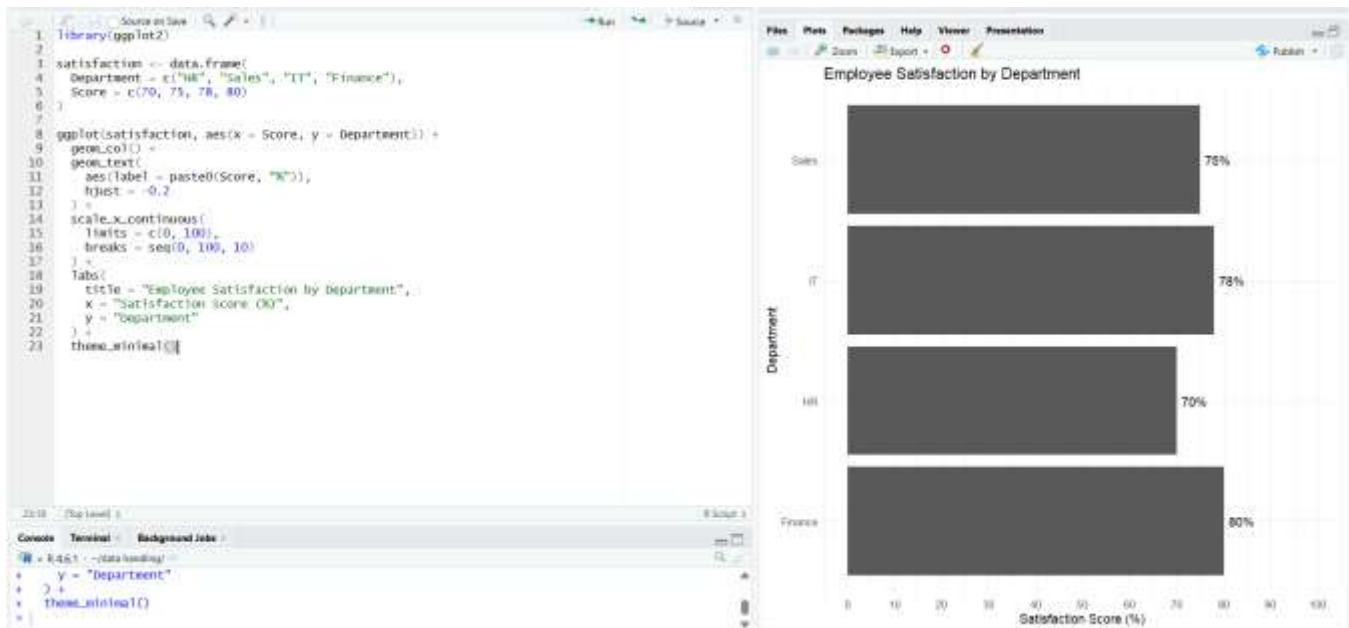
Horizontal Bar Chart with a 0–100% Axis

A horizontal bar chart is suitable because it clearly compares satisfaction scores across different departments. Starting the X-axis at 0% prevents the differences from appearing exaggerated, and the exact scores can be easily read from the bars.

Procedure:

1. Open RStudio and load the ggplot2 package.
2. Create a data frame containing departments and satisfaction scores.
3. Use `geom_col()` to create proportional bars.
4. Set the X-axis from **0% to 100%** to avoid exaggerating differences.
5. Add percentage labels and use a simple accessible color scheme without red-green coding.

Output:



Justification:

The original visualization is misleading because its Y-axis starts at 65%, making the 70–80% differences appear much larger than they actually are. Starting the axis at 0% follows the proportional ink principle and accurately represents bar lengths. Removing red-green coding improves accessibility for users with color-vision deficiencies. The redesigned graph shows that Finance has the highest score (80%) and HR the lowest (70%), a difference of only 10 percentage points.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)		Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____